

LAIKA's Next Generation Stage Planning System

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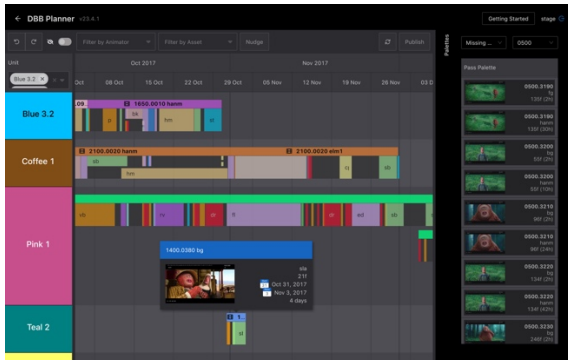


Figure 1: (Left) DBB Web Planner. (Right) Tabletop Planner 1.0 live-updates DBB Display.

ABSTRACT

LAIKA launched the Digital Big Boards (DBB) Web Planner project to scale up production and increase accuracy in the stage shooting schedule. This is the first major update since 2017, when LAIKA launched the DBB Planner, a native application built for custom tabletop touch displays that replaced the traditional analog method of pinning task cards onto physical boards. The DBB suite schedules the stage production timeline and renders live updates, a breakthrough that converted the shooting schedule into accessible and up-to-date information for the whole studio. Its success spawned spin-off applications for stage support crew and asset fabricators to manage their daily schedules. However, after adoption, we found scalability and usability issues with the tabletop DBB Planner. Difficulty making real-time updates via the touchscreen app, along with interdependent apps requiring increased accuracy in daily scheduling led to the development of its successor: the Web Planner. This new web application improves real-time performance and scales to support larger concurrent productions.

CCS CONCEPTS

• Human-centered computing; • Web-based interaction; • Software and its engineering; • Software evolution;

KEYWORDS

scheduling, productivity, animation feature film production

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1 INTRODUCTION

1.1 Evolution of stage scheduling at LAIKA

Stage scheduling at LAIKA is a complex process that requires careful planning across many departments. Stop-motion animation requires all physical assets (sets, props, puppets, faces, costumes) to be ready at shoot time, and coordination between departments to ensure that stages are ready for an animator. This shooting schedule is represented as *big boards*: large displays that provide a visual representation of the production timeline of tasks.

Stop-motion is a very tactile process, and traditionally big boards were as well: giant pin boards displayed the entirety of the film's shots, each task represented as a pinned note per stage per day. Delays in the shooting schedule required crew members to painstakingly shift tasks by hand throughout the course of the show, which could not be relied upon for fine-tuned accuracy.

We replaced this manual process with the *Digital Big Boards* (DBB) [Stringer et al., 2017], which scheduled half of *Kubo and the Two Strings* and the entirety of our last feature, *Missing Link*. DBB scheduled with $\frac{1}{4}$ day accuracy and directed the delivery of thousands of fabricated assets. DBB Planner 1.0 created a studio-wide view of the schedule that was more accurate, up-to-date, and easier to maintain. The digital schedule instructed priorities across the fabrication departments and the stage support teams, and we deployed a suite of stage support apps that rely on the schedule to support these departments' workflows.

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1.2 Limitations of DBB Planner 1.0

DBB Planner 1.0 has not scaled to meet LAIKA's growing needs and the rising complexity of our productions. The current show has nearly double the number of shots of any previous show (~2900 shots compared to 1635 shots in *Missing Link*), and DBB 1.0 was unable to scale to support the hundreds of thousands of tasks of the new production.

The goal for DBB Planner 1.0 had been to provide a scheduling experience as similar as possible to the analog big boards, intentionally taking a skeuomorphic approach to drive rapid adoption with minimal training requirements. The app ran on 55" touchscreen tabletop displays. Scheduling was handled via drag-and-drop interactions akin to the physical shuffling of shot cards and hand-pinning of notes onto a board. Planner 1.0 pulled in multiple years' worth of tasks scheduled across >50 shooting stages. The tactile nature of its design helped early adoption by the crew but led to usability difficulties later. Daily adjustments were difficult to manage through touch interaction and drag-and-drop resource scheduling was comparably slow and cumbersome compared to bulk task editing within ShotGrid. Failure to capture necessary shifts in the schedule would impact the fabrication departments and stage support crew downstream, whose tooling both relied on and fed into the stage production timeline. This created new urgency and new usability requirements for managing an even more detailed and faster-changing schedule.

Since the launch of DBB, new production crew have joined the team that are unfamiliar with the analog scheduling methods that the original Planner sought to mimic. They are comfortable in a desktop setting, and as such, we abandoned skeuomorphism to create a new scheduling application: the DBB Web Planner. Accessible via a standard web browser, schedulers are no longer tied to dedicated hardware and can work anywhere, enabling more schedulers to work concurrently onsite or remotely.

2 DESIGN

The Web Planner architecture consists of three layers: (1) a ShotGrid API service, (2) a data store that caches the entire timeline and users' planning sessions, and (3) a web client that allows end users to manipulate the schedule.

We were able to reuse the ShotGrid API service from the predecessor Planner 1.0, which handles reading and publishing tasks to the production database. DBB Planner 1.0 fetched all tasks for each active show and stored them in-process for fast access. While this worked well for *Missing Link*, this system did not scale for more ambitious projects and does not work for a multi-show setting.

To overcome these limitations, we developed a new backend service written in Rust that caches tasks from ShotGrid and stores planning sessions in Redis. It is responsible for caching the entirety of the stage production timeline while exposing a much smaller subset of tasks to web clients over GraphQL. Complex schedule computations happen server-side in the data store layer, and we write interim schedule data to the Redis store, until we publish to ShotGrid and rebuild the cache. Our architecture is designed to reduce the load on ShotGrid as well as improve the response time to the client application, bringing initial load times down from several minutes to seconds.

The client layer is responsible for rendering the user interface: a scheduler view that allows drag-and-drop interactions to schedule tasks onto stages, assign resources, and make bulk-task adjustments. We used the Bryntum suite of scheduling tools to build the UI. Bryntum provides powerful scheduling visualization libraries with scalable, performant task rendering for large data sets. Users can filter and search by resource (crew and assets), which can have customized calendars to convey availability. Resource collisions are now visible at-a-glance, while the predecessor Planner required custom reporting to analyze resource availability. We use a multi-scheduler view to represent planning scenarios, where users can select a subset of tasks, plan out adjustments to the schedule, and visualize its impact without updating the production timeline.

3 FUTURE WORK

We intend to leverage AI-powered automatic resource-leveling using Autodesk's Generative Scheduling feature. This would automate scheduling given the complex series of cross-departmental resource constraints and delivery dates involved in the shoot. LAIKA is an early development partner with Autodesk on this feature. We currently use it to automate resource leveling within our fabrication departments [Nowakowski et al., 2020].

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